PART V

Ideals and Factor Rings

Section 26 Homomorphisms and Factor Rings
Section 27 Prime and Maximal Ideals
Section 28 †Gröbner Bases for Ideals

SECTION 26 HOMOMORPHISMS AND FACTOR RINGS

Homomorphisms

We defined the concepts of *homomorphism* and *isomorphism* for rings in Section 18, since we wished to talk about evaluation homomorphisms for polynomials and about isomorphic rings. We repeat some definitions here for easy reference. Recall that a homomorphism is a *structure-relating map*. A homomorphism for rings must relate both their additive structure and their multiplicative structure.

26.1 Definition A map ϕ of a ring R into a ring R' is a **homomorphism** if

$$\phi(a + b) = \phi(a) + \phi(b)$$

and

$$\phi(ab) = \phi(a)\phi(b)$$

for all elements a and b in R. ■

In Example 18.10 we defined evaluation homomorphisms, and Example 18.11 showed that the map $\phi : \mathbb{Z} \to \mathbb{Z}_n$, where $\phi(m)$ is the remainder of m when divided by n, is a homomorphism. We give another simple but very fundamental example of a homomorphism.

26.2 Example **(Projection Homomorphisms)** Let $R_1, R_2, \cdots, R_n$ be rings. For each i, the map $\pi_i : R_1 \times R_2 \times \cdots \times R_n \to R_i$ defined by $\pi_i(r_1, r_2, \cdots, r_n) = r_i$ is a homomorphism, *projection onto the ith component*. The two required properties of a homomorphism hold

† Section 28 is not required for the remainder of the text.

for π_i since both addition and multiplication in the direct product are computed by addition and multiplication in each individual component. ▲

Properties of Homomorphisms

We work our way through the exposition of Section 13 but for ring homomorphisms.

26.3 Theorem **(Analogue of Theorem 13.12)** Let ϕ be a homomorphism of a ring R into a ring R'. If 0 is the additive identity in R, then $\phi(0) = 0'$ is the additive identity in R', and if $a \in R$, then $\phi(-a) = -\phi(a)$. If S is a subring of R, then $\phi[S]$ is a subring of R'. Going the other way, if S' is a subring of R', then $\phi^{-1}[S']$ is a subring of R. Finally, if R has unity 1, then $\phi(1)$ is unity for $\phi[R]$. Loosely speaking, subrings correspond to subrings, and rings with unity correspond to rings with unity under a ring homomorphism.

Proof Let ϕ be a homomorphism of a ring R into a ring R'. Since, in particular, ϕ can be viewed as a group homomorphism of $\langle R, +\rangle$ into $\langle R', +'\rangle$, Theorem 13.12 tells us that $\phi(0) = 0'$ is the additive identity element of R' and that $\phi(-a) = -\phi(a)$.

Theorem 13.12 also tells us that if S is a subring of R, then, considering the additive group $\langle S, +\rangle$, the set $\langle \phi[S], +'\rangle$ gives a subgroup of $\langle R', +'\rangle$. If $\phi(s_1)$ and $\phi(s_2)$ are two elements of $\phi[S]$, then

$$\phi(s_1)\phi(s_2) = \phi(s_1 s_2)$$

and $\phi(s_1 s_2) \in \phi[S]$. Thus $\phi(s_1)\phi(s_2) \in \phi[S]$, so $\phi[S]$ is closed under multiplication. Consequently, $\phi[S]$ is a subring of R'.

Going the other way, Theorem 13.12 also shows that if S' is a subring of R', then $\langle \phi^{-1}[S'], +\rangle$ is a subgroup of $\langle R, +\rangle$. Let $a, b \in \phi^{-1}[S']$, so that $\phi(a) \in S'$ and $\phi(b) \in S'$. Then

$$\phi(ab) = \phi(a)\phi(b).$$

Since $\phi(a)\phi(b) \in S'$, we see that $ab \in \phi^{-1}[S']$ so $\phi^{-1}[S']$ is closed under multiplication and thus is a subring of R.

Finally, if R has unity 1, then for all $r \in R$,

$$\phi(r) = \phi(1r) = \phi(r1) = \phi(1)\phi(r) = \phi(r)\phi(1),$$

so $\phi(1)$ is unity for $\phi[R]$. ◆

Note in Theorem 26.3 that $\phi(1)$ is unity for $\phi[R]$, but not necessarily for R' as we ask you to illustrate in Exercise 9.

26.4 Definition Let a map $\phi : R \to R$ be a homomorphism of rings. The subring

$$\phi^{-1}[0'] = \{r \in R \mid \phi(r) = 0'\}$$

is the **kernel** of ϕ, denoted by $\text{Ker}(\phi)$. ■

Now this $\text{Ker}(\phi)$ is the same as the kernel of the group homomorphism of $\langle R, +\rangle$ into $\langle R', +\rangle$ given by ϕ. Theorem 13.15 and Corollary 13.18 on group homomorphisms give us at once analogous results for ring homomorphisms.

26.5 Theorem **(Analogue of Theorem 13.15)** Let $\phi : R \to R'$ be a ring homomorphism, and let $H = \text{Ker}(\phi)$. Let $a \in R$. Then $\phi^{-1}[\phi(a)] = a + H = H + a$, where $a + H = H + a$ is the coset containing a of the commutative additive group $\langle H, + \rangle$.

26.6 Corollary **(Analogue of Corollary 13.18)** A ring homomorphism $\phi : R \to R'$ is a one-to-one map if and only if $\text{Ker}(\phi) = \{0\}$.

Factor (Quotient) Rings

We are now ready to describe the analogue for rings of Section 14. We start with the analogue of Theorem 14.1.

26.7 Theorem **(Analogue of Theorem 14.1)** Let $\phi : R \to R'$ be a ring homomorphism with kernel H. Then the additive cosets of H form a ring R/H whose binary operations are defined by choosing representatives. That is, the sum of two cosets is defined by

$$(a + H) + (b + H) = (a + b) + H,$$

and the product of the cosets is defined by

$$(a + H)(b + H) = (ab) + H.$$

Also, the map $\mu : R/H \to \phi[R]$ defined by $\mu(a + H) = \phi(a)$ is an isomorphism.

Proof Once again, the additive part of the theory is done for us in Theorem 14.1. We proceed to check the multiplicative aspects.

We must first show that multiplication of cosets by choosing representatives is well defined. To this end, let $h_1, h_2, \in H$ and consider the representatives $a + h_1$ of $a + H$ and $b + h_2$ of $b + H$. Let

$$c = (a + h_1)(b + h_2) = ab + ah_2 + h_1b + h_1h_2.$$

We must show that this element c lies in the coset $ab + H$. Since $ab + H = \phi^{-1}[\phi(ab)]$, we need only show that $\phi(c) = \phi(ab)$. Since ϕ is a homomorphism and $\phi(h) = 0'$ for $h \in H$, we obtain

$$\begin{aligned} \phi(c) &= \phi(ab + ah_2 + h_1b + h_1h_2) \\ &= \phi(ab) + \phi(ah_2) + \phi(h_1b) + \phi(h_1h_2) \\ &= \phi(ab) + \phi(a)0' + 0'\phi(b) + 0'0' \\ &= \phi(ab) + 0' + 0' + 0' = \phi(ab). \end{aligned} \tag{1}$$

Thus multiplication by choosing representatives is well defined.

To show that R/H is a ring, it remains to show that the associative property for multiplication and the distributive laws hold in R/H. Since addition and multiplication are computed by choosing representatives, these properties follow at once from corresponding properties in R.

Theorem 14.1 shows that the map μ defined in the statement of Theorem 26.4 is well defined, one to one, onto $\phi[R]$, and satisfies the additive property for a homomorphism.

Multiplicatively, we have

$$\mu[(a+H)(b+H)] = \mu(ab+H) = \phi(ab)$$
$$= \phi(a)\phi(b) = \mu(a+H)\mu(b+H).$$

This completes the demonstration that μ is an isomorphism. ◆

26.8 Example Example 18.11 shows that the map $\phi : \mathbb{Z} \to \mathbb{Z}_n$ defined by $\phi(m) = r$, where r is the remainder of m when divided by n, is a homomorphism. Since $\text{Ker}(\phi) = n\mathbb{Z}$, Theorem 26.7 shows that $\mathbb{Z}/n\mathbb{Z}$ is a ring where operations on residue classes can be computed by choosing representatives and performing the corresponding operation in $\mathbb{Z}$. The theorem also shows that this ring $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. ▲

It remains only to characterize those subrings H of a ring R such that multiplication of additive cosets of H by choosing representatives is well defined. The coset multiplication in Theorem 26.7 was shown to be well defined in Eq. (1). The success of Eq. (1) is due to the fact that $\phi(ah_2) = \phi(h_1 b) = \phi(h_1 h_2) = 0'$. That is, if $h \in H$ where $H = \text{Ker}(\phi)$, then for every $a, b \in R$ we have $ah \in H$ and $hb \in H$. This suggests Theorem 26.9 below, which is the analogue of Theorem 14.4.

26.9 Theorem **(Analogue of Theorem 14.4)** Let H be a subring of the ring R. Multiplication of additive cosets of H is well defined by the equation

$$(a+H)(b+H) = ab+H$$

if and only if $ah \in H$ and $hb \in H$ for all $a, b \in R$ and $h \in H$.

Proof Suppose first that $ah \in H$ and $hb \in H$ for all $a, b \in R$ and all $h \in H$. Let $h_1, h_2 \in H$ so that $a + h_1$ and $b + h_2$ are also representatives of the cosets $a + H$ and $b + H$ containing a and b. Then

$$(a+h_1)(b+h_2) = ab + ah_2 + h_1 b + h_1 h_2.$$

Since ah_2 and $h_1 b$ and $h_1 h_2$ are all in H by hypothesis, we see that $(a + h_1)(b + h_2) \in ab + H$.

Conversely, suppose that multiplication of additive cosets by representatives is well defined. Let $a \in R$ and consider the coset product $(a + H)H$. Choosing representatives $a \in (a + H)$ and $0 \in H$, we see that $(a + H)H = a0 + H = 0 + H = H$. Since we can also compute $(a + H)H$ by choosing $a \in (a + H)$ and any $h \in H$, we see that $ah \in H$ for any $h \in H$. A similar argument starting with the product $H(b + H)$ shows that $hb \in H$ for any $h \in H$. ◆

In group theory, normal subgroups are precisely the type of substructure of groups required to form a factor group with a well-defined operation on cosets given by operating with chosen representatives. Theorem 26.9 shows that in ring theory, the analogous substructure must be a subring H of a ring R such that $aH \subseteq H$ and $Hb \subseteq H$ for all $a, b \in R$, where $aH = \{ah \mid h \in H\}$ and $Hb = \{hb \mid h \in H\}$. From now on we will usually denote such a substructure by N rather than H. Recall that we started using N to mean a normal subgroup in Section 15.

26.10 Definition An addive subgroup N of a ring R satisfying the properties

$$aN \subseteq N \quad \text{and} \quad Nb \subseteq N \quad \text{for all } a, b \in R$$

is an **ideal.** ■

26.11 Example We see that $n\mathbb{Z}$ is an ideal in the ring $\mathbb{Z}$ since we know it is a subring, and $s(nm) = (nm)s = n(ms) \in n\mathbb{Z}$ for all $s \in \mathbb{Z}$. ▲

26.12 Example Let F be the ring of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let C be the subring of F consisting of all the constant functions in F. Is C an ideal in F? Why?

Solution It is not true that the product of a constant function with every function is again a constant function. For example, the product of $\sin x$ and 2 is the function $2 \sin x$. Thus C is not an ideal of F. ▲

■ Historical Note

It was Ernst Eduard Kummer (1810–1893) who introduced the concept of an "ideal complex number" in 1847 in order to preserve the notion of unique factorization in certain rings of algebraic integers. In particular, Kummer wanted to be able to factor into primes numbers of the form $a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{p-1}\alpha^{p-1}$, where α is a complex root of $x^p = 1$ (p prime) and the a_i are ordinary integers. Kummer had noticed that the naive definition of primes as "unfactorable numbers" does not lead to the expected results; the product of two such "unfactorable" numbers may well be divisible by other "unfactorable" numbers. Kummer defined "ideal prime factors" and "ideal numbers" in terms of certain congruence relationships; these "ideal factors" were then used as the divisors necessary to preserve unique factorization. By use of these, Kummer was in fact able to prove certain cases of Fermat's Last Theorem, which states that $x^n + y^n = z^n$ has no solutions $x, y, z \in \mathbb{Z}^+$ if $n > 2$.

It turned out that an "ideal number," which was in general not a "number" at all, was uniquely determined by the set of integers it "divided." Richard Dedekind took advantage of this fact to identify the ideal factor with this set; he therefore called the set itself an ideal, and proceeded to show that it satisfied the definition given in the text. Dedekind was then able to define the notions of prime ideal and product of two ideals and show that any ideal in the ring of integers of any algebraic number field could be written uniquely as a product of prime ideals.

26.13 Example Let F be as in the preceding example, and let N be the subring of all functions f such that $f(2) = 0$. Is N an ideal in F? Why or why not?

Solution Let $f \in N$ and let $g \in F$. Then $(fg)(2) = f(2)g(2) = 0g(2) = 0$, so $fg \in N$. Similarly, we find that $gf \in N$. Therefore N is an ideal of F. We could also have proved this by just observing that N is the kernel of the evaluation homomorphism $\phi_2 : F \to \mathbb{R}$. ▲

Once we know that multiplication by choosing representatives is well defined on additive cosets of a subring N of R, the associative law for multiplication and the distributive laws for these cosets follow at once from the same properties in R. We have at once this corollary of Theorem 26.9.

26.14 Corollary **(Analogue of Corollary 14.5)** Let N be an ideal of a ring R. Then the additive cosets of N form a ring R/N with the binary operations defined by

$$(a + N) + (b + N) = (a + b) + N$$

and

$$(a + N)(b + N) = ab + N.$$

26.15 Definition The ring R/N in the preceding corollary is the **factor ring** (or **quotient ring**) **of** R **by** N. ■

If we use the term *quotient ring*, be sure not to confuse it with the notion of the *field of quotients* of an integral domain, discussed in Section 21.

Fundamental Homomorphism Theorem

To complete our analogy with Sections 13 and 14, we give the analogues of Theorems 14.9 and 14.11.

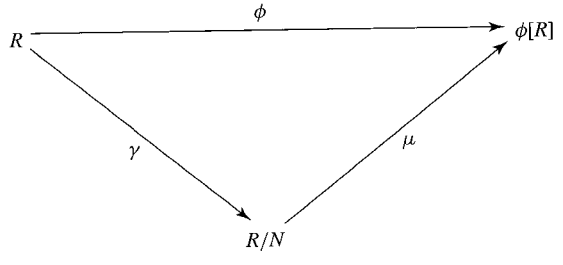
26.16 Theorem **(Analogue of Theorem 14.9)** Let N be an ideal of a ring R. Then $\gamma : R \to R/N$ given by $\gamma(x) = x + N$ is a ring homomorphism with kernel N.

Proof The additive part is done in Theorem 14.9. Turning to the multiplicative question, we see that

$$\gamma(xy) = (xy) + N = (x + N)(y + N) = \gamma(x)\gamma(y).$$ ◆

26.17 Theorem **(Fundamental Homomorphism Theorem; Analogue of Theorem 14.11)** Let $\phi : R \to R'$ be a ring homomorphism with kernel N. Then $\phi[R]$ is a ring, and the map $\mu : R/N \to \phi[R]$ given by $\mu(x + N) = \phi(x)$ is an isomorphism. If $\gamma : R \to R/N$ is the homomorphism given by $\gamma(x) = x + N$, then for each $x \in R$, we have $\phi(x) = \mu\gamma(x)$.

Proof This follows at once from Theorems 26.7 and 26.16. Figure 26.18 is the analogue of Fig. 14.10. ◆



26.18 Figure

26.19 Example Example 26.11 shows that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$, so we can form the factor ring $\mathbb{Z}/n\mathbb{Z}$. Example 18.11 shows that $\phi : \mathbb{Z} \to \mathbb{Z}_n$ where $\phi(m)$ is the remainder of m modulo n is a homomorphism, and we see that $\text{Ker}(\phi) = n\mathbb{Z}$. Theorem 26.17 then shows that the map $\mu : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}_n$ where $\mu(m + n\mathbb{Z})$ is the remainder of m modulo n is well defined and is an isomorphism. ▲

In summary, every ring homomorphism with domain R gives rise to a factor ring R/N, and every factor ring R/N gives rise to a homomorphism mapping R into R/N. An *ideal* in ring theory is analogous to a *normal subgroup* in the group theory. Both are the type of substructure needed to form a factor structure.

We should now add an addendum to Theorem 26.3 on properties of homomorphisms. Let $\phi : R \to R'$ be a homomorphism, and let N be an ideal of R. Then $\phi[N]$ is an ideal of $\phi[R]$, although it need not be an ideal of R'. Also, if N' is an ideal of either $\phi[R]$ or of R', then $\phi^{-1}[N']$ is an ideal of R. We leave the proof of this to Exercise 22.

EXERCISES 26

Computations

1. Describe all ring homomorphisms of $\mathbb{Z} \times \mathbb{Z}$ into $\mathbb{Z} \times \mathbb{Z}$. [*Hint:* Note that if ϕ is such a homomorphism, then $\phi((1, 0)) = \phi((1, 0))\phi((1, 0))$ and $\phi((0, 1)) = \phi((0, 1))\phi((0, 1))$. Consider also $\phi((1, 0)(0, 1))$.]
2. Find all positive integers n such that $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$.
3. Find all ideals N of $\mathbb{Z}_{12}$. In each case compute $\mathbb{Z}_{12}/N$; that is, find a known ring to which the quotient ring is isomorphic.
4. Give addition and multiplication tables for $2\mathbb{Z}/8\mathbb{Z}$. Are $2\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}_4$ isomorphic rings?

Concepts

In Exercises 5 through 7, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

5. An *isomorphism of a ring* R with a ring R' is a homomorphism $\phi : R \to R'$ such that $\text{Ker}(\phi) = \{0\}$.
6. An *ideal* N of a ring R is an additive subgroup of $\langle R, + \rangle$ such that for all $r \in R$ and all $n \in N$, we have $rn \in N$ and $nr \in N$.
7. The *kernel of a homomorphism* ϕ mapping a ring R into a ring R' is $\{\phi(r) = 0' \mid r \in R\}$.
8. Let F be the ring of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ and having derivatives of all orders. Differentiation gives a map $\delta : F \to F$ where $\delta(f(x)) = f'(x)$. Is δ a homomorphism? Why? Give the connection between this exercise and Example 26.12.
9. Give an example of a ring homomorphism $\phi : R \to R'$ where R has unity 1 and $\phi(1) \neq 0'$, but $\phi(1)$ is not unity for R'.
10. Mark each of the following true or false.

    _____ **a.** The concept of a ring homomorphism is closely connected with the idea of a factor ring.

    _____ **b.** A ring homomorphism $\phi : R \to R'$ carries ideals of R into ideals of R'.

    _____ **c.** A ring homomorphism is one to one if and only if the kernel is $\{0\}$.

    _____ **d.** $\mathbb{Q}$ is an ideal in $\mathbb{R}$.

_____ **e.** Every ideal in a ring is a subring of the ring.

_____ **f.** Every subring of every ring is an ideal of the ring.

_____ **g.** Every quotient ring of every commutative ring is again a commutative ring.

_____ **h.** The rings $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}_4$ are isomorphic.

_____ **i.** An ideal N in a ring R with unity 1 is all of R if and only if $1 \in N$.

_____ **j.** The concept of an ideal is to the concept of a ring as the concept of a normal subgroup is to the concept of a group.

11. Let R be a ring. Observe that $\{0\}$ and R are both ideals of R. Are the factor rings R/R and $R/\{0\}$ of real interest? Why?

12. Give an example to show that a factor ring of an integral domain may be a field.

13. Give an example to show that a factor ring of an integral domain may have divisors of 0.

14. Give an example to show that a factor ring of a ring with divisors of 0 may be an integral domain.

15. Find a subring of the ring $\mathbb{Z} \times \mathbb{Z}$ that is not an ideal of $\mathbb{Z} \times \mathbb{Z}$.

16. A student is asked to prove that a quotient ring of a ring R modulo an ideal N is commutative if and only if $(rs - sr) \in N$ for all $r, s \in R$. The student starts out:
Assume R/N is commutative. Then $rs = sr$ for all $r, s \in R/N$.

a. Why does the instructor reading this expect nonsense from there on?

b. What should the student have written?

c. Prove the assertion. (Note the "if and only if.")

Theory

17. Let $R = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ and let R' consist of all 2×2 matrices of the form $\begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{Z}$. Show that R is a subring of $\mathbb{R}$ and that R' is a subring of $M_2(\mathbb{Z})$. Then show that $\phi : R \to R'$, where $\phi(a + b\sqrt{2}) = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ is an isomorphism.

18. Show that each homomorphism from a field to a ring is either one to one or maps everything onto 0.

19. Show that if R, R', and R'' are rings, and if $\phi : R \to R'$ and $\psi : R' \to R''$ are homomorphisms, then the composite function $\psi\phi : R \to R''$ is a homomorphism. (Use Exercise 49 of Section 13.)

20. Let R be a commutative ring with unity of prime characteristic p. Show that the map $\phi_p : R \to R$ given by $\phi_p(a) = a^p$ is a homomorphism (the **Frobenius homomorphism**).

21. Let R and R' be rings and let $\phi : R \to R'$ be a ring homomorphism such that $\phi[R] \neq \{0'\}$. Show that if R has unity 1 and R' has no 0 divisors, then $\phi(1)$ is unity for R'.

22. Let $\phi : R \to R'$ be a ring homomorphism and let N be an ideal of R.

a. Show that $\phi[N]$ is an ideal of $\phi[R]$.

b. Give an example to show that $\phi[N]$ need not be an ideal of R'.

c. Let N' be an ideal either of $\phi[R]$ or of R'. Show that $\phi^{-1}[N']$ is an ideal of R.

23. Let F be a field, and let S be any subset of $F \times F \times \cdots \times F$ for n factors. Show that the set N_S of all $f(x_1, \cdots, x_n) \in F[x_1, \cdots, x_n]$ that have every element $(a_1, \cdots, a_n)$ of S as a zero (see Exercise 28 of Section 22) is an ideal in $F[x_1, \cdots, x_n]$. This is of importance in algebraic geometry.

24. Show that a factor ring of a field is either the trivial (zero) ring of one element or is isomorphic to the field.

25. Show that if R is a ring with unity and N is an ideal of R such that $N \neq R$, then R/N is a ring with unity.

26. Let R be a commutative ring and let $a \in R$. Show that $I_a = \{x \in R \mid ax = 0\}$ is an ideal of R.

27. Show that an intersection of ideals of a ring R is again an ideal of R.

28. Let R and R' be rings and let N and N' be ideals of R and R', respectively. Let ϕ be a homomorphism of R into R'. Show that ϕ induces a natural homomorphism $\phi_* : R/N \to R'/N'$ if $\phi[N] \subseteq N'$. (Use Exercise 39 of Section 14.)

29. Let ϕ be a homomorphism of a ring R with unity onto a nonzero ring R'. Let u be a unit in R. Show that $\phi(u)$ is a unit in R'.

30. An element a of a ring R is **nilpotent** if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that the collection of all nilpotent elements in a commutative ring R is an ideal, the **nilradical of** R.

31. Referring to the definition given in Exercise 30, find the nilradical of the ring $\mathbb{Z}_{12}$ and observe that it is one of the ideals of $\mathbb{Z}_{12}$ found in Exercise 3. What is the nilradical of $\mathbb{Z}$? of $\mathbb{Z}_{32}$?

32. Referring to Exercise 30, show that if N is the nilradical of a commutative ring R, then R/N has as nilradical the trivial ideal $\{0 + N\}$.

33. Let R be a commutative ring and N an ideal of R. Referring to Exercise 30, show that if every element of N is nilpotent and the nilradical of R/N is R/N, then the nilradical of R is R.

34. Let R be a commutative ring and N an ideal of R. Show that the set $\sqrt{N}$ of all $a \in R$, such that $a^n \in N$ for some $n \in \mathbb{Z}^+$, is an ideal of R, the **radical of** N.

35. Referring to Exercise 34, show by examples that for proper ideals N of a commutative ring R,

a. $\sqrt{N}$ need not equal N **b.** $\sqrt{N}$ may equal N.

36. What is the relationship of the ideal $\sqrt{N}$ of Exercise 34 to the nilradical of R/N (see Exercise 30)? Word your answer carefully.

37. Show that $\phi : \mathbb{C} \to M_2(\mathbb{R})$ given by

$$\phi(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

for $a, b \in \mathbb{R}$ gives an isomorphism of $\mathbb{C}$ with the subring $\phi[\mathbb{C}]$ of $M_2(\mathbb{R})$.

38. Let R be a ring with unity and let $\text{End}(\langle R, + \rangle)$ be the ring of endomorphisms of $\langle R, + \rangle$ as described in Section 24. Let $a \in R$, and let $\lambda_a : R \to R$ be given by

$$\lambda_a(x) = ax$$

for $x \in R$.

a. Show that λ_a is an endomorphism of $\langle R, + \rangle$.

b. Show that $R' = \{\lambda_a \mid a \in R\}$ is a subring of $\text{End}(\langle R, + \rangle)$.

c. Prove the analogue of Cayley's theorem for R by showing that R' of (b) is isomorphic to R.

SECTION 27 Prime and Maximal Ideals

Exercises 12 through 14 of the preceding section asked us to provide examples of factor rings R/N where R and R/N have very different structural properties. We start with some examples of this situation, and in the process, provide solutions to those exercises.

27.1 Example As was shown in Corollary 19.12, the ring $\mathbb{Z}_p$, which is isomorphic to $\mathbb{Z}/p\mathbb{Z}$, is a field for p a prime. *Thus a factor ring of an integral domain may be a field.* ▲

27.2 Example The ring $\mathbb{Z} \times \mathbb{Z}$ is not an integral domain, for

$$(0, 1)(1, 0) = (0, 0),$$

showing that $(0, 1)$ and $(1, 0)$ are 0 divisors. Let $N = \{(0, n) \mid n \in \mathbb{Z}\}$. Now N is an ideal of $\mathbb{Z} \times \mathbb{Z}$, and $(\mathbb{Z} \times \mathbb{Z})/N$ is isomorphic to $\mathbb{Z}$ under the correspondence $[(m, 0) + N] \leftrightarrow m$, where $m \in \mathbb{Z}$. Thus a *factor* ring of a ring may be an integral domain, even though the original ring is not. ▲

27.3 Example The subset $N = \{0, 3\}$ of $\mathbb{Z}_6$ is easily seen to be an ideal of $\mathbb{Z}_6$, and $\mathbb{Z}_6/N$ has three elements, $0 + N$, $1 + N$, and $2 + N$. These add and multiply in such a fashion as to show that $\mathbb{Z}_6/N \simeq \mathbb{Z}_3$ under the correspondence

$$(0 + N) \leftrightarrow 0, \qquad (1 + N) \leftrightarrow 1, \qquad (2 + N) \leftrightarrow 2.$$

This example shows that *if R is not even an integral domain, that is, if R has zero divisors, it is still possible for R/N to be a field.* ▲

27.4 Example Note that $\mathbb{Z}$ is an integral domain, but $\mathbb{Z}/6\mathbb{Z} \simeq \mathbb{Z}_6$ is not. The preceding examples showed that a factor ring may have a structure that seems *better* than the original ring. This example indicates that the structure of a factor ring may seem *worse* than that of the original ring. ▲

Every nonzero ring R has at least two ideals, the **improper ideal** R and the **trivial ideal** $\{0\}$. For these ideals, the factor rings are R/R, which has only one element, and $R/\{0\}$, which is isomorphic to R. These are uninteresting cases. Just as for a subgroup of a group, a **proper nontrivial ideal** of a ring R is an ideal N of R such that $N \neq R$ and $N \neq \{0\}$.

While factor rings of rings and integral domains may be of great interest, as the above examples indicate, Corollary 27.6, which follows our next theorem, shows that a factor ring of a field is really not useful to us.

27.5 Theorem If R is a ring with unity, and N is an ideal of R containing a unit, then $N = R$.

Proof Let N be an ideal of R, and suppose that $u \in N$ for some unit u in R. Then the condition $rN \subseteq N$ for all $r \in R$ implies, if we take $r = u^{-1}$ and $u \in N$, that $1 = u^{-1}u$ is in N. But then $rN \subseteq N$ for all $r \in R$ implies that $r1 = r$ is in N for all $r \in R$, so $N = R$. ◆

27.6 Corollary A field contains no proper nontrivial ideals.

Proof Since every nonzero element of a field is a unit, it follows at once from Theorem 27.5 that an ideal of a field F is either $\{0\}$ or all of F. ◆

Maximal and Prime Ideals

We now take up the question of when a factor ring is a field and when it is an integral domain. The analogy with groups in Section 15 can be stretched a bit further to cover the case in which the factor ring is a field.

27.7 Definition A **maximal ideal of a ring** R is an ideal M different from R such that there is no proper ideal N of R properly containing M. ■

27.8 Example Let p be a prime positive integer. We know that $\mathbb{Z}/p\mathbb{Z}$ is isomorphic to $\mathbb{Z}_p$. Forgetting about multiplication for the moment and regarding $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Z}_p$ as additive groups, we know that $\mathbb{Z}_p$ is a simple group, and consequently $p\mathbb{Z}$ must be a maximal normal subgroup of $\mathbb{Z}$ by Theorem 15.18. Since $\mathbb{Z}$ is an abelian group and every subgroup is a normal subgroup, we see that $p\mathbb{Z}$ is a maximal proper subgroup of $\mathbb{Z}$. Since $p\mathbb{Z}$ is an ideal of the ring $\mathbb{Z}$, it follows that $p\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$. We know that $\mathbb{Z}/p\mathbb{Z}$ is isomorphic to the ring $\mathbb{Z}_p$, and that $\mathbb{Z}_p$ is actually a field. Thus $\mathbb{Z}/p\mathbb{Z}$ is a field. This illustrates the next theorem. ▲

27.9 Theorem **(Analogue of Theorem 15.18)** Let R be a commutative ring with unity. Then M is a maximal ideal of R if and only if R/M is a field.

Proof Suppose M is a maximal ideal in R. Observe that if R is a commutative ring with unity, then R/M is also a nonzero commutative ring with unity if $M \neq R$, which is the case if M is maximal. Let $(a + M) \in R/M$, with $a \notin M$, so that $a + M$ is not the additive identity element of R/M. Suppose $a + M$ has no multiplicative inverse in R/M. Then the set $(R/M)(a + M) = \{(r + M)(a + M) \mid (r + M) \in R/M\}$ does not contain $1 + M$. We easily see that $(R/M)(a + M)$ is an ideal of R/M. It is nontrivial because $a \notin M$, and it is a proper ideal because it does not contain $1 + M$. By the final paragraph of Section 26, if $\gamma : R \to R/M$ is the canonical homomorphism, then $\gamma^{-1}[(R/M)(a + M)]$ is a proper ideal of R properly containing M. But this contradicts our assumption that M is a maximal ideal, so $a + M$ must have a multiplicative inverse in R/M.

Conversely, suppose that R/M is a field. By the final paragraph of Section 26, if N is any ideal of R such that $M \subset N \subset R$ and γ is the canonical homomorphism of R onto R/M, then $\gamma[N]$ is an ideal of R/M with $\{(0 + M)\} \subset \gamma[N] \subset R/M$. But this is contrary to Corollary 27.6, which states that the field R/M contains no proper nontrivial ideals. Hence if R/M is a field, M is maximal. ◆

27.10 Example Since $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$ and $\mathbb{Z}_n$ is a field if and only if n is a prime, we see that the maximal ideals of $\mathbb{Z}$ are precisely the ideals $p\mathbb{Z}$ for prime positive integers p. ▲

27.11 Corollary A commutative ring with unity is a field if and only if it has no proper nontrivial ideals.

Proof Corollary 27.6 shows that a field has no proper nontrivial ideals.

Conversely, if a commutative ring R with unity has no proper nontrivial ideals, then $\{0\}$ is a maximal ideal and $R/\{0\}$, which is isomorphic to R, is a field by Theorem 27.9. ◆

We now turn to the question of characterizing, for a commutative ring R with unity, the ideals $N \neq R$ such that R/N is an integral domain. The answer here is rather obvious. The factor ring R/N will be an integral domain if and only if $(a + N)(b + N) = N$ implies that either

$$a + N = N \qquad \text{or} \qquad b + N = N.$$

This is exactly the statement that R/N has no divisors of 0, since the coset N plays the role of 0 in R/N. Looking at representatives, we see that this condition amounts to saying that $ab \in N$ implies that either $a \in N$ or $b \in N$.

27.12 Example All ideals of $\mathbb{Z}$ are of the form $n\mathbb{Z}$. For $n = 0$, we have $n\mathbb{Z} = \{0\}$, and $\mathbb{Z}/\{0\} \simeq \mathbb{Z}$, which is an integral domain. For $n > 0$, we have $\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}_n$ and $\mathbb{Z}_n$ is an integral domain if and only if n is a prime. Thus the nonzero ideals $n\mathbb{Z}$ such that $\mathbb{Z}/n\mathbb{Z}$ is an integral domain are of the form $p\mathbb{Z}$, where p is a prime. Of course, $\mathbb{Z}/p\mathbb{Z}$ is actually a field, so that $p\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$. Note that for a product rs of integers to be in $p\mathbb{Z}$, the prime p must divide either r or s. The role of prime integers in this example makes the use of the word *prime* in the next definition more reasonable. ▲

27.13 Definition An ideal $N \neq R$ in a commutative ring R is a **prime ideal** if $ab \in N$ implies that either $a \in N$ or $b \in N$ for $a, b \in R$. ■

Note that $\{0\}$ is a prime ideal in $\mathbb{Z}$, and indeed, in any integral domain.

27.14 Example Note that $\mathbb{Z} \times \{0\}$ is a prime ideal of $\mathbb{Z} \times \mathbb{Z}$, for if $(a, b)(c, d) \in \mathbb{Z} \times \{0\}$, then we must have $bd = 0$ in $\mathbb{Z}$. This implies that either $b = 0$ so $(a, b) \in \mathbb{Z} \times \{0\}$ or $d = 0$ so $(c, d) \in \mathbb{Z} \times \{0\}$. Note that $(\mathbb{Z} \times \mathbb{Z})/(\mathbb{Z} \times \{0\})$ is isomorphic to $\mathbb{Z}$, which is an integral domain. ▲

Our remarks preceding Example 27.12 constitute a proof of the following theorem, which is illustrated by Example 27.14.

27.15 Theorem Let R be a commutative ring with unity, and let $N \neq R$ be an ideal in R. Then R/N is an integral domain if and only if N is a prime ideal in R.

27.16 Corollary Every maximal ideal in a commutative ring R with unity is a prime ideal.

Proof If M is maximal in R, then R/M is a field, hence an integral domain, and therefore M is a prime ideal by Theorem 27.15. ◆

The material that has just been presented regarding maximal and prime ideals is very important and we shall be using it quite a lot. We should keep the main ideas well in mind. We must know and understand the definitions of maximal and prime ideals and must remember the following facts that we have demonstrated.

For a commutative ring R with unity:

1. An ideal M of R is maximal if and only if R/M is a field.
2. An ideal N of R is prime if and only if R/N is an integral domain.
3. Every maximal ideal of R is a prime ideal.

Prime Fields

We now proceed to show that the rings $\mathbb{Z}$ and $\mathbb{Z}_n$ form foundations upon which all rings with unity rest, and that $\mathbb{Q}$ and $\mathbb{Z}_p$ perform a similar service for all fields. Let R be any ring with unity 1. Recall that by $n \cdot 1$ we mean $1 + 1 + \cdots + 1$ for n summands for $n > 0$, and $(-1) + (-1) + \cdots + (-1)$ for $|n|$ summands for $n < 0$, while $n \cdot 1 = 0$ for $n = 0$.

27.17 Theorem If R is a ring with unity 1, then the map $\phi : \mathbb{Z} \to R$ given by

$$\phi(n) = n \cdot 1$$

for $n \in \mathbb{Z}$ is a homomorphism of $\mathbb{Z}$ into R.

Proof Observe that

$$\phi(n + m) = (n + m) \cdot 1 = (n \cdot 1) + (m \cdot 1) = \phi(n) + \phi(m).$$

The distributive laws in R show that

$$\underbrace{(1 + 1 + \cdots + 1)}_{n \text{ summands}}\underbrace{(1 + 1 + \cdots + 1)}_{m \text{ summands}} = \underbrace{(1 + 1 + \cdots + 1)}_{nm \text{ summands}}.$$

Thus $(n \cdot 1)(m \cdot 1) = (nm) \cdot 1$ for $n, m > 0$. Similar arguments with the distributive laws show that for all $n, m \in \mathbb{Z}$, we have

$$(n \cdot 1)(m \cdot 1) = (nm) \cdot 1.$$

Thus

$$\phi(nm) = (nm) \cdot 1 = (n \cdot 1)(m \cdot 1) = \phi(n)\phi(m). \quad \blacklozenge$$

27.18 Corollary If R is a ring with unity and characteristic $n > 1$, then R contains a subring isomorphic to $\mathbb{Z}_n$. If R has characteristic 0, then R contains a subring isomorphic to $\mathbb{Z}$.

Proof The map $\phi : \mathbb{Z} \to R$ given by $\phi(m) = m \cdot 1$ for $m \in \mathbb{Z}$ is a homomorphism by Theorem 27.17. The kernel must be an ideal in $\mathbb{Z}$. All ideals in $\mathbb{Z}$ are of the form $s\mathbb{Z}$ for some $s \in \mathbb{Z}$. By Theorem 19.15 we see that if R has characteristic $n > 0$, then the kernel of ϕ is $n\mathbb{Z}$. Then the image $\phi[\mathbb{Z}] \leq R$ is isomorphic to $\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}_n$. If the characteristic of R is 0, then $m \cdot 1 \neq 0$ for all $m \neq 0$, so the kernel of ϕ is $\{0\}$. Thus, the image $\phi[\mathbb{Z}] \leq R$ is isomorphic to $\mathbb{Z}$. $\blacklozenge$

27.19 Theorem A field F is either of prime characteristic p and contains a subfield isomorphic to $\mathbb{Z}_p$ or of characteristic 0 and contains a subfield isomorphic to $\mathbb{Q}$.

Proof If the characteristic of F is not 0, the above corollary shows that F contains a subring isomorphic to $\mathbb{Z}_n$. Then n must be a prime p, or F would have 0 divisors. If F is of characteristic 0, then F must contain a subring isomorphic to $\mathbb{Z}$. In this case Corollaries

21.8 and 21.9 show that F must contain a field of quotients of this subring and that this field of quotients must be isomorphic to $\mathbb{Q}$. ◆

Thus every field contains either a subfield isomorphic to $\mathbb{Z}_p$ for some prime p or a subfield isomorphic to $\mathbb{Q}$. These fields $\mathbb{Z}_p$ and $\mathbb{Q}$ are the fundamental building blocks on which all fields rest.

27.20 Definition The fields $\mathbb{Z}_p$ and $\mathbb{Q}$ are **prime fields.** ■

Ideal Structure in $F[x]$

Throughout the rest of this section, we assume that F is a field. We give the next definition for a general commutative ring R with unity, although we are only interested in the case $R = F[x]$. Note that for a commutative ring R with unity and $a \in R$, the set $\{ra \mid r \in R\}$ is an ideal in R that contains the element a.

27.21 Definition If R is a commutative ring with unity and $a \in R$, the ideal $\{ra \mid r \in R\}$ of all multiples of a is the **principal ideal generated by** a and is denoted by $\langle a \rangle$. An ideal N of R is a **principal ideal** if $N = \langle a \rangle$ for some $a \in R$. ■

27.22 Example Every ideal of the ring $\mathbb{Z}$ is of the form $n\mathbb{Z}$, which is generated by n, so every ideal of $\mathbb{Z}$ is a principal ideal. ▲

27.23 Example The ideal $\langle x \rangle$ in $F[x]$ consists of all polynomials in $F[x]$ having zero constant term. ▲

The next theorem is another simple but very important application of the division algorithm for $F[x]$. (See Theorem 23.1.) The proof of this theorem is to the division algorithm in $F[x]$ as the proof that a subgroup of a cyclic group is cyclic is to the division algorithm in $\mathbb{Z}$.

27.24 Theorem If F is a field, every ideal in $F[x]$ is principal.

Proof Let N be an ideal of $F[x]$. If $N = \{0\}$, then $N = \langle 0 \rangle$. Suppose that $N \neq \{0\}$, and let $g(x)$ be a nonzero element of N of minimal degree. If the degree of $g(x)$ is 0, then $g(x) \in F$ and is a unit, so $N = F[x] = \langle 1 \rangle$ by Theorem 27.5, so N is principal. If the degree of $g(x)$ is ≥ 1, let $f(x)$ be any element of N. Then by Theorem 23.1, $f(x) = g(x)q(x) + r(x)$, where $r(x) = 0$ or (degree $r(x)$) $<$ (degree $g(x)$). Now $f(x) \in N$ and $g(x) \in N$ imply that $f(x) - g(x)q(x) = r(x)$ is in N by definition of an ideal. Since $g(x)$ is a nonzero element of minimal degree in N, we must have $r(x) = 0$. Thus $f(x) = g(x)q(x)$ and $N = \langle g(x) \rangle$. ◆

We can now characterize the maximal ideals of $F[x]$. This is a crucial step in achieving our **basic goal:** to show that any nonconstant polynomial $f(x)$ in $F[x]$ has a zero in some field E containing F.

27.25 Theorem An ideal $\langle p(x)\rangle \neq \{0\}$ of $F[x]$ is maximal if and only if $p(x)$ is irreducible over F.

Proof Suppose that $\langle p(x)\rangle \neq \{0\}$ is a maximal ideal of $F[x]$. Then $\langle p(x)\rangle \neq F[x]$, so $p(x) \notin F$. Let $p(x) = f(x)g(x)$ be a factorization of $p(x)$ in $F[x]$. Since $\langle p(x)\rangle$ is a maximal ideal and hence also a prime ideal, $(f(x)g(x)) \in \langle p(x)\rangle$ implies that $f(x) \in \langle p(x)\rangle$ or $g(x) \in \langle p(x)\rangle$; that is, either $f(x)$ or $g(x)$ has $p(x)$ as a factor. But then we can't have the degrees of both $f(x)$ and $g(x)$ less than the degree of $p(x)$. This shows that $p(x)$ is irreducible over F.

Conversely, if $p(x)$ is irreducible over F, suppose that N is an ideal such that $\langle p(x)\rangle \subseteq N \subseteq F[x]$. Now N is a principal ideal by Theorem 27.24, so $N = \langle g(x)\rangle$ for some $g(x) \in N$. Then $p(x) \in N$ implies that $p(x) = g(x)q(x)$ for some $q(x) \in F[x]$. But $p(x)$ is irreducible, which implies that either $g(x)$ or $q(x)$ is of degree 0. If $g(x)$ is of degree 0, that is, a nonzero constant in F, then $g(x)$ is a unit in $F[x]$, so $\langle g(x)\rangle = N = F[x]$. If $q(x)$ is of degree 0, then $q(x) = c$, where $c \in F$, and $g(x) = (1/c)p(x)$ is in $\langle p(x)\rangle$, so $N = \langle p(x)\rangle$. Thus $\langle p(x)\rangle \subset N \subset F[x]$ is impossible, so $\langle p(x)\rangle$ is maximal. ◆

27.26 Example Example 23.9 shows that $x^3 + 3x + 2$ is irreducible in $\mathbb{Z}_5[x]$, so $\mathbb{Z}_5[x]/\langle x^3 + 3x + 2\rangle$ is a field. Similarly, Theorem 22.11 shows that $x^2 - 2$ is irreducible in $\mathbb{Q}[x]$, so $\mathbb{Q}[x]/\langle x^2 - 2\rangle$ is a field. We shall examine such fields in more detail later. ▲

Application to Unique Factorization in $F[x]$

In Section 23, we stated without proof Theorem 27.27, which follows. (See Theorem 23.18.) Assuming this theorem, we proved in Section 23 that factorization of polynomials in $F[x]$ into irreducible polynomials is unique, except for order of factors and units in F. We delayed the proof of Theorem 27.27 until now since the machinery we have developed enables us to give such a simple, four-line proof. This proof fills the gap in our proof of unique factorization in $F[x]$.

27.27 Theorem Let $p(x)$ be an irreducible polynomial in $F[x]$. If $p(x)$ divides $r(x)s(x)$ for $r(x), s(x) \in F[x]$, then either $p(x)$ divides $r(x)$ or $p(x)$ divides $s(x)$.

Proof Suppose $p(x)$ divides $r(x)s(x)$. Then $r(x)s(x) \in \langle p(x)\rangle$, which is maximal by Theorem 27.25. Therefore, $\langle p(x)\rangle$ is a prime ideal by Corollary 27.16. Hence $r(x)s(x) \in \langle p(x)\rangle$ implies that either $r(x) \in \langle p(x)\rangle$, giving $p(x)$ divides $r(x)$, or that $s(x) \in \langle p(x)\rangle$, giving $p(x)$ divides $s(x)$. ◆

A Preview of Our Basic Goal

We close this section with an outline of the demonstration in Section 29 of our basic goal. We have all the ideas for the proof at hand now; perhaps you can fill in the details from this outline.

Basic goal: Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Show that there exists a field E containing F and containing a zero α of $f(x)$.

Outline of the Proof

1. Let $p(x)$ be an irreducible factor of $f(x)$ in $F[x]$.
2. Let E be the *field* $F[x]/\langle p(x)\rangle$. (See Theorems 27.25 and 27.9.)
3. Show that no two different elements of F are in the same coset of $F[x]/\langle p(x)\rangle$, and deduce that we may consider F to be (isomorphic to) a subfield of E.
4. Let α be the coset $x + \langle p(x)\rangle$ in E. Show that for the evaluation homomorphism $\phi_\alpha : F[x] \to E$, we have $\phi_\alpha(f(x)) = 0$. That is, α is a zero of $f(x)$ in E.

An example of a field constructed according to this outline is given in Section 29. There, we give addition and multiplication tables for the field $\mathbb{Z}_2[x]/\langle x^2 + x + 1\rangle$. We show there that this field has just four elements, the cosets

$$0 + \langle x^2 + x + 1\rangle, \qquad 1 + \langle x^2 + x + 1\rangle, \qquad x + \langle x^2 + x + 1\rangle,$$

and

$$(x + 1) + \langle x^2 + x + 1\rangle.$$

We rename these four cosets 0, 1, α, and $\alpha + 1$ respectively, and obtain Tables 29.20 and 29.21 for addition and multiplication in this 4-element field. To see how these tables are constructed, remember that we are in a field of characteristic 2, so that $\alpha + \alpha = \alpha(1 + 1) = \alpha 0 = 0$. Remember also that α is a zero of $x^2 + x + 1$, so that $\alpha^2 + \alpha + 1 = 0$ and consequently $\alpha^2 = -\alpha - 1 = \alpha + 1$.

EXERCISES 27

Computations

1. Find all prime ideals and all maximal ideals of $\mathbb{Z}_6$.
2. Find all prime ideals and all maximal ideals of $\mathbb{Z}_{12}$.
3. Find all prime ideals and all maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_2$.
4. Find all prime ideals and all maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$.
5. Find all $c \in \mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/\langle x^2 + c\rangle$ is a field.
6. Find all $c \in \mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/\langle x^3 + x^2 + c\rangle$ is a field.
7. Find all $c \in \mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/\langle x^3 + cx^2 + 1\rangle$ is a field.
8. Find all $c \in \mathbb{Z}_5$ such that $\mathbb{Z}_5[x]/\langle x^2 + x + c\rangle$ is a field.
9. Find all $c \in \mathbb{Z}_5$ such that $\mathbb{Z}_5[x]/\langle x^2 + cx + 1\rangle$ is a field.

Concepts

In Exercises 10 through 13, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

10. A *maximal ideal* of a ring R is an ideal that is not contained in any other ideal of R.
11. A *prime ideal* of a commutative ring R is an ideal of the form $pR = \{pr \mid r \in R\}$ for some prime p.

12. A *prime field* is a field that has no proper subfields.

13. A *principal ideal* of a commutative ring with unity is an ideal N with the property that there exists $a \in N$ such that N is the smallest ideal that contains a.

14. Mark each of the following true or false.

_____ **a.** Every prime ideal of every commutative ring with unity is a maximal ideal.
_____ **b.** Every maximal ideal of every commutative ring with unity is a prime ideal.
_____ **c.** $\mathbb{Q}$ is its own prime subfield.
_____ **d.** The prime subfield of $\mathbb{C}$ is $\mathbb{R}$.
_____ **e.** Every field contains a subfield isomorphic to a prime field.
_____ **f.** A ring with zero divisors may contain one of the prime fields as a subring.
_____ **g.** Every field of characteristic zero contains a subfield isomorphic to $\mathbb{Q}$.
_____ **h.** Let F be a field. Since $F[x]$ has no divisors of 0, every ideal of $F[x]$ is a prime ideal.
_____ **i.** Let F be a field. Every ideal of $F[x]$ is a principal ideal.
_____ **j.** Let F be a field. Every principal ideal of $F[x]$ is a maximal ideal.

15. Find a maximal ideal of $\mathbb{Z} \times \mathbb{Z}$.

16. Find a prime ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not maximal.

17. Find a nontrivial proper ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not prime.

18. Is $\mathbb{Q}[x]/\langle x^2 - 5x + 6\rangle$ a field? Why?

19. Is $\mathbb{Q}[x]/\langle x^2 - 6x + 6\rangle$ a field? Why?

Proof Synopsis

20. Give a one- or two-sentence synopsis of "only if" part of Theorem 27.9.

21. Give a one- or two-sentence synopsis of "if" part of Theorem 27.9.

22. Give a one- or two-sentence synopsis of Theorem 27.24.

23. Give a one- or two-sentence synopsis of the "only if" part of Theorem 27.25.

Theory

24. Let R be a finite commutative ring with unity. Show that every prime ideal in R is a maximal ideal.

25. Corollary 27.18 tells us that every ring with unity contains a subring isomorphic to either $\mathbb{Z}$ or some $\mathbb{Z}_n$. Is it possible that a ring with unity may simultaneously contain two subrings isomorphic to $\mathbb{Z}_n$ and $\mathbb{Z}_m$ for $n \neq m$? If it is possible, give an example. If it is impossible, prove it.

26. Continuing Exercise 25, is it possible that a ring with unity may simultaneously contain two subrings isomorphic to the fields $\mathbb{Z}_p$ and $\mathbb{Z}_q$ for two different primes p and q? Give an example or prove it is impossible.

27. Following the idea of Exercise 26, is it possible for an integral domain to contain two subrings isomorphic to $\mathbb{Z}_p$ and $\mathbb{Z}_q$ for $p \neq q$ and p and q both prime? Give reasons or an illustration.

28. Prove directly from the definitions of maximal and prime ideals that every maximal ideal of a commutative ring R with unity is a prime ideal. [*Hint*: Suppose M is maximal in R, $ab \in M$, and $a \notin M$. Argue that the smallest ideal $\{ra + m \mid r \in R, m \in M\}$ containing a and M must contain 1. Express 1 as $ra + m$ and multiply by b.]

29. Show that N is a maximal ideal in a ring R if and only if R/N is a **simple ring,** that is, it is nontrivial and has no proper nontrivial ideals. (Compare with Theorem 15.18.)

30. Prove that if F is a field, every proper nontrivial prime ideal of $F[x]$ is maximal.

31. Let F be a field and $f(x), g(x) \in F[x]$. Show that $f(x)$ divides $g(x)$ if and only if $g(x) \in \langle f(x) \rangle$.

32. Let F be a field and let $f(x), g(x) \in F[x]$. Show that

$$N = \{r(x)f(x) + s(x)g(x) \mid r(x), s(x) \in F[x]\}$$

is an ideal of $F[x]$. Show that if $f(x)$ and $g(x)$ have different degrees and $N \neq F[x]$, then $f(x)$ and $g(x)$ cannot both be irreducible over F.

33. Use Theorem 27.24 to prove the *equivalence* of these two theorems:

Fundamental Theorem of Algebra: Every nonconstant polynomial in $\mathbb{C}[x]$ has a zero in $\mathbb{C}$.

Nullstellensatz for $\mathbb{C}[x]$: Let $f_1(x), \cdots, f_r(x) \in \mathbb{C}[x]$ and suppose that every $\alpha \in \mathbb{C}$ that is a zero of all r of these polynomials is also a zero of a polynomial $g(x)$ in $\mathbb{C}[x]$. Then some power of $g(x)$ is in the smallest ideal of $\mathbb{C}[x]$ that contains the r polynomials $f_1(x), \cdots, f_r(x)$.

There is a sort of arithmetic of ideals in a ring. The next three exercises define sum, product, and quotient of ideals.

34. If A and B are ideals of a ring R, the **sum $A + B$ of A and B** is defined by

$$A + B = \{a + b \mid a \in A, b \in B\}.$$

a. Show that $A + B$ is an ideal. **b.** Show that $A \subseteq A + B$ and $B \subseteq A + B$.

35. Let A and B be ideals of a ring R. The **product AB of A and B** is defined by

$$AB = \left\{ \sum_{i=1}^{n} a_i b_i \mid a_i \in A, b_i \in B, n \in \mathbb{Z}^+ \right\}.$$

a. Show that AB is an ideal in R. **b.** Show that $AB \subseteq (A \cap B)$.

36. Let A and B be ideals of a *commutative* ring R. The **quotient $A : B$ of A by B** is defined by

$$A : B = \{r \in R \mid rb \in A \text{ for all } b \in B\}.$$

Show that $A : B$ is an ideal of R.

37. Show that for a field F, the set S of all matrices of the form

$$\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$$

for $a, b \in F$ is a **right ideal** but not a **left ideal** of $M_2(F)$. That is, show that S is a subring closed under multiplication on the *right* by any element of $M_2(F)$, but is not closed under *left* multiplication.

38. Show that the matrix ring $M_2(\mathbb{Z}_2)$ is a simple ring; that is, $M_2(\mathbb{Z}_2)$ has no proper nontrivial ideals.

SECTION 28 †GRÖBNER BASES FOR IDEALS

This section gives a brief introduction to algebraic geometry. In particular, we are concerned with the problem of finding as simple a description as we can of the set of common zeros of a finite number of polynomials. In order to accomplish our goal in a single section of this text, we will be stating a few theorems without proof. We recommend the book by Adams and Loustaunau [23] for the proofs and further study.

† This section is not used in the remainder of the text.

Algebraic Varieties and Ideals

Let F be a field. Recall that $F[x_1, x_2, \cdots, x_n]$ is the ring of polynomials in n indeterminants $x_1, x_2, \cdots, x_n$ with coefficients in F. We let F^n be the Cartesian product $F \times F \times \cdots \times F$ for n factors. For ease in writing, we denote an element $(a_1, a_2, \cdots, a_n)$ of F^n by **a,** in bold type. Using similar economy, we let $F[\mathbf{x}] = F[x_1, x_2, \cdots, x_n]$. For each $\mathbf{a} \in F^n$, we have an evaluation homomorphism $\phi_{\mathbf{a}}: F[\mathbf{x}] \to F$ just as in Theorem 22.4. That is, for $f(\mathbf{x}) = f(x_1, x_2, \cdots, x_n) \in F[\mathbf{x}]$, we define $\phi_{\mathbf{a}}(f(\mathbf{x})) = f(\mathbf{a}) = f(a_1, a_2, \cdots, a_n)$. The proof that $\phi_{\mathbf{a}}$ is indeed a homomorphism follows from the associative, commutative, and distributive properties of the operations in $F[\mathbf{x}]$ and F. Just as for the one-indeterminate case, an element **a** of F^n is a **zero of** $f(\mathbf{x}) \in F[\mathbf{x}]$ if $f(\mathbf{a}) = 0$. In what follows, we further abbreviate a polynomial $f(\mathbf{x})$ by "f."

In this section we discuss the problem of finding common zeros in F^n of a finite number of polynomials $f_1, f_2, \cdots, f_r$ in $F[\mathbf{x}]$. Finding and studying geometric properties of the set of all these common zeros is the subject of algebraic geometry.

28.1 Definition Let S be a finite subset of $F[\mathbf{x}]$. The **algebraic variety** $V(S)$ in F^n is the set of all common zeros in F^n of the polynomials in S. ■

In our illustrative examples, which usually involve at most three indeterminates, we use x, y, z in place of x_1, x_2, and x_3.

28.2 Example Let $S = \{2x + y - 2\} \subset \mathbb{R}[x, y]$. The algebraic variety $V(S)$ in $\mathbb{R}^2$ is the line with x-intercept 1 and y-intercept 2. ▲

We leave to Exercise 29 the straightforward proof that for r elements $f_1, f_2, \cdots, f_r$ in a commutative ring R with unity, the set

$$I = \{c_1 f_1 + c_2 f_2 + \cdots + c_r f_r \mid c_i \in R \text{ for } i = 1, \cdots, r\}$$

is an ideal of R. We denote this ideal by $\langle f_1, f_2, \cdots, f_r \rangle$. We are interested in the case $R = F[\mathbf{x}]$ where all the c_i and all the f_i are polynomials in $F[\mathbf{x}]$. We regard the c_i as "coefficient polynomials." By its construction, this ideal I is the smallest ideal containing the polynomials $f_1, f_2, \cdots, f_r$; it can also be described as the intersection of all ideals containing these r polynomials.

28.3 Definition Let I be an ideal in a commutative ring R with unity. A subset $\{b_1, b_2, \cdots, b_r\}$ of I is a **basis** for I if $I = \langle b_1, b_2, \cdots, b_r \rangle$. ■

Unlike the situation in linear algebra, there is no requirement of independence for elements of a basis, or of unique representation of an ideal member in terms of a basis.

28.4 Theorem Let $f_1, f_2, \cdots, f_r \in F[\mathbf{x}]$. The set of common zeros in F^n of the polynomials f_i for $i = 1, 2, \cdots, r$ is the same as the set of common zeros in F^n of all the polynomials in the entire ideal $I = \langle f_1, f_2, \cdots, f_r \rangle$.

Proof Let

$$f = c_1 f_1 + c_2 f_2 + \cdots + c_r f_r \tag{1}$$

be any element of I, and let $\mathbf{a} \in F^n$ be a common zero of $f_1, f_2, \cdots$, and f_r. Applying the evaluation homomorphism $\phi_{\mathbf{a}}$ to Eq. (1), we obtain

$$\begin{aligned} f(\mathbf{a}) &= c_1(\mathbf{a})f_1(\mathbf{a}) + c_2(\mathbf{a})f_2(\mathbf{a}) + \cdots + c_r(\mathbf{a})f_r(\mathbf{a}) \\ &= c_1(\mathbf{a})0 + c_2(\mathbf{a})0 + \cdots + c_r(\mathbf{a})0 = 0, \end{aligned}$$

showing that $\mathbf{a}$ is also a zero of every polynomial f in I. Of course, a zero of every polynomial in I will be a zero of each f_i because each $f_i \in I$. ◆

For an ideal I in $F[\mathbf{x}]$, we let $V(I)$ be the set of all common zeros of all elements of I. We can summarize Theorem 28.4 as

$$V(\{f_1, f_2, \cdots, f_r\}) = V(\langle f_1, f_2, \cdots, f_r\rangle).$$

We state without proof the Hilbert Basis Theorem. (See Adams and Loustaunau [23].)

28.5 Theorem **(Hilbert Basis Theorem)** Every ideal in $F[x_1, x_2, \cdots, x_n]$ has a finite basis.

> **Our objective**: Given a basis for an ideal I in $F[\mathbf{x}]$, modify it if possible to become a basis that better exhibits the structure of I and the geometry of the associated algebraic variety $V(I)$.

The theorem that follows provides a tool for this task. You should notice that the theorem gives information about the division algorithm that we did not mention in Theorem 23.1. We use the same notation here as in Theorem 23.1, but with $\mathbf{x}$ rather than x. If $f(\mathbf{x}) = g(\mathbf{x})h(\mathbf{x})$ in $F(\mathbf{x})$, then $g(\mathbf{x})$ and $h[\mathbf{x}]$ are called **"divisors"** or **"factors"** of $f(\mathbf{x})$.

28.6 Theorem **(Property of the Division Algorithm)** Let $f(\mathbf{x})$, $g(\mathbf{x})$, $q(\mathbf{x})$ and $r(\mathbf{x})$ be polynomials in $F[\mathbf{x}]$ such that $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$. The common zeros in F^n of $f(\mathbf{x})$ and $g(\mathbf{x})$ are the same as the common zeros of $g(\mathbf{x})$ and $r(\mathbf{x})$. Also the common divisors in $F[\mathbf{x}]$ of $f(\mathbf{x})$ and $g(\mathbf{x})$ are the same as the common divisors of $g(\mathbf{x})$ and $r(\mathbf{x})$.

If $f(\mathbf{x})$ and $g(\mathbf{x})$ are two members of a basis for an ideal I of $F[\mathbf{x}]$, then replacement of $f(\mathbf{x})$ by $r(\mathbf{x})$ in the basis still yields a basis for I.

Proof If $\mathbf{a} \in F^n$ is a common zero of $g(\mathbf{x})$ and $r(\mathbf{x})$, then applying $\phi_{\mathbf{a}}$ to both sides of the equation $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$, we obtain $f(\mathbf{a}) = g(\mathbf{a})q(\mathbf{a}) + r(\mathbf{a}) = 0q(\mathbf{a}) + 0 = 0$, so $\mathbf{a}$ is a zero of both $f(\mathbf{x})$ and $g(\mathbf{x})$. If $\mathbf{b} \in F[\mathbf{x}]$ is a common zero of $f(\mathbf{x})$ and $g(\mathbf{x})$, then applying $\phi_{\mathbf{b}}$ yields $f(\mathbf{b}) = g(\mathbf{b})q(\mathbf{b}) + r(\mathbf{b})$ so $0 = 0q(\mathbf{b}) + r(\mathbf{b})$ and we see that $r(\mathbf{b}) = 0$ as well as $g(\mathbf{b})$.

The proof concerning common divisors is essentially the same, and is left as Exercise 30.

Finally, let B be a basis for an ideal I, let $f(\mathbf{x})$, $g(\mathbf{x})$, $\in B$ and let $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$. Let B' be the set obtained by replacing $f(\mathbf{x})$ by $r(\mathbf{x})$ in B, and let I' be the ideal having B' as a basis. Let S be the set obtained from B by adjoining $r(\mathbf{x})$ to B. Note that S can also be obtained by adjoining $f(\mathbf{x})$ to B'. The equation $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$

shows that $f(\mathbf{x}) \in I'$, so we have $B' \subseteq S \subseteq I'$. Thus S is a basis for I'. The equation $r(\mathbf{x}) = f(\mathbf{x}) - q(\mathbf{x})g(\mathbf{x})$ shows that $r(\mathbf{x}) \in I$, so we have $B \subseteq S \subseteq I$. Thus S is basis for I. Therefore $I = I'$ and B' is a basis for I. ◆

A Familiar Linear Illustration

A basic technique for problem solving in linear algebra is finding all common solutions of a finite number of linear equations. For the moment we abandon our practice of never writing "$f(\mathbf{x}) = 0$" for a nonzero polynomial, and work a typical problem as we do in a linear algebra course.

28.7 Example **(Solution as in a Linear Algebra Course)** Find all solutions in $\mathbb{R}^3$ of the linear system

$$\begin{aligned} x + y - 3z &= 8 \\ 2x + y + z &= -5. \end{aligned}$$

Solution We multiply the first equation by -2 and add it to the second, obtaining the new system

$$\begin{aligned} x + y - 3z &= 8 \\ -y + 7z &= -21 \end{aligned}$$

which has the same solution set in $\mathbb{R}^3$ as the preceding one. For any value z, we can find the corresponding y-value from the second equation and then determine x from the first equation. Keeping z as parameter, we obtain $\{(-4z - 13, 7z + 21, z) \mid z \in \mathbb{R}\}$ as solution set, which is a line in Euclidean 3-space through the point $(-13, 21, 0)$. ▲

In the notation of this section, the problem in the preceding example can be phrased as follows:

$$\text{Describe } V(\langle x + y - 3z - 8, 2x + y + z + 5\rangle) \text{ in } \mathbb{R}^3.$$

We solved it by finding a more useful basis, namely

$$\{x + y - 3z - 8, -y + 7z + 21\}.$$

Notice that the second member, $-y + 7z + 21$, of this new basis can be obtained from the original two basis polynomials as a remainder $r(x, y, z)$ in a division process, namely

$$\begin{array}{r} 2 \\ x + y - 3z - 8 \overline{\big|\; 2x + y + z + 5} \\ \underline{2x + 2y - 6z - 16} \\ -y + 7z + 21 \end{array}$$

Thus $2x + y + z + 5 = (x + y - 3z - 8)(2) + (-y + 7z + 21)$, an expression of the form $f(x, y, z) = g(x, y, z)q(x, y, z) + r(x, y, z)$. We replaced the polynomial f by the polynomial r, as in Theorem 28.6, which assures us that $V(\langle f, g\rangle) = V(\langle g, r\rangle)$ and that $\langle f, g\rangle = \langle g, r\rangle$. We chose a very simple, 1-step problem in Example 28.7. However, it is clear that the method introduced in a linear algebra course for solving a linear system can be phrased in terms of applying a division algorithm process repeatedly to change a given ideal basis into one that better illuminates the geometry of the associated algebraic variety.

A Single Indeterminate Illustration

Suppose now that we want to find the variety $V(I)$ in $\mathbb{R}$ associated with an ideal I in $F[x]$, the ring of polynomials in the single indeterminate x. By Theorem 27.24, every ideal in $F[x]$ is principal, so there exists $f(x) \in F[x]$ such that $I = \langle f(x) \rangle$. Thus $V(I)$ consists of the zeros of a single polynomial, and $\{f(x)\}$ is probably as simple a basis for I as we could desire. We give an example illustrating computation of such a single generator $f(x)$ for I in a case where the given basis for I contains more than one polynomial. Because a polynomial in $\mathbb{R}[x]$ has only a finite number of zeros in $\mathbb{R}$, we expect two or more randomly selected polynomials in $\mathbb{R}[x]$ to have no common zeros, but we constructed the basis in our example carefully!

28.8 Example Let us describe the algebraic variety V in $\mathbb{R}$ consisting of common zeros of

$$f(x) = x^4 + x^3 - 3x^2 - 5x - 2 \quad \text{and} \quad g(x) = x^3 + 3x^2 - 6x - 8.$$

We want to find a new basis for $\langle f, g \rangle$ having polynomials of as small degree as possible, so we use the division algorithm $f(x) = g(x)q(x) + r(x)$ in Theorem 23.1, where $r(x)$ will have degree at most 2. We then replace the basis $\{f, g\}$ by the basis $\{g, r\}$.

$$\begin{array}{r|l} & x - 2 \\ \hline x^3 + 3x^2 - 6x - 8 & x^4 + x^3 - 3x^2 - 5x - 2 \\ & \underline{x^4 + 3x^3 - 6x^2 - 8x} \\ & \quad -2x^3 + 3x^2 + 3x - 2 \\ & \quad \underline{-2x^3 - 6x^2 + 12x + 16} \\ & \quad\quad 9x^2 - 9x - 18 \end{array}$$

Because zeros of $9x^2 - 9x - 18$ are the same as zeros of $x^2 - x - 2$, we let $r(x) = x^2 - x - 2$, and take as new basis

$$\{g, r\} = (x^3 + 3x^2 - 6x - 8, x^2 - x - 2).$$

By dividing $g(x)$ by $r(x)$ to obtain a remainder $r_1(x)$, we will now be able to find a basis $\{r(x), r_1(x)\}$ consisting of polynomials of degree at most 2.

$$\begin{array}{r|l} & x + 4 \\ \hline x^2 - x - 2 & x^3 + 3x^2 - 6x - 8 \\ & \underline{x^3 - x^2 - 2x} \\ & \quad 4x^2 - 4x - 8 \\ & \quad \underline{4x^2 - 4x - 8} \\ & \quad\quad 0 \end{array}$$

Our new basis $\{r(x), r_1(x)\}$ now becomes $\{x^2 - x - 2\}$. Thus $I = \langle f(x), g(x) \rangle = \langle x^2 - x - 2 \rangle = \langle (x - 2)(x + 1) \rangle$, and we see that $V = \{-1, 2\}$. ▲

Theorem 28.6 tells us that the common divisors of $f(x)$ and $g(x)$ in the preceding example are the same as the common divisors of $r(x)$ and $r_1(x)$. Because $0 = (0)r(x)$, we see that $r(x)$ itself divides 0, so the common divisors of $f(x)$ and $g(x)$ are just those of $r(x)$, which, of course, include $r(x)$ itself. Thus $r(x)$ is called a "*greatest common divisor*" (abbreviated gcd) of $f(x)$ and $g(x)$.

Gröbner Bases

We tackle the problem of finding a nice basis for an ideal I in $F[\mathbf{x}] = F[x_1, x_2, \cdots, x_n]$. In view of our illustrations for the linear and single indeterminant cases, it seems reasonable to try to replace polynomials in a basis by polynomials of lower degree, or containing fewer indeterminates. It is crucial to have a systematic way to accomplish this. Every instructor in linear algebra has had an occasional student who refuses to master matrix reduction and creates zero entries in columns of a matrix in an almost random fashion, rather than finishing the first column and then proceeding to the second, etc. As a first step in our goal, we tackle this problem of specifying an order for polynomials in a basis.

Our polynomials in $F[\mathbf{x}]$ have terms of the form $ax_1^{m_1}x_2^{m_2}\cdots x_n^{m_n}$ where $a \in F$.

Properties for an Ordering of Power Products

1. $1 < P$ for all power products $P \neq 1$.
2. For any two power products P_i and P_j, exactly one of $P_i < P_j$, $P_i = P_j$, $P_j < P_i$ holds.
3. If $P_i < P_j$ and $P_j < P_k$, then $P_i < P_k$.
4. If $P_i < P_j$, then $PP_i < PP_j$ for any power product P.

Let us consider a **power product** in $F[\mathbf{x}]$ to be an expression

$$P = x_1^{m_1}x_2^{m_2}\cdots x_n^{m_n} \text{ where all the } m_i \geq 0 \text{ in } \mathbb{Z}.$$

Notice that all x_i are present, perhaps some with exponent 0. Thus in $F[x, y, z]$, we must write xz^2 as xy^0z^2 to be a power product. We want to describe a *total ordering* $<$ on the set of all power products so that we know just what it means to say that $P_i < P_j$ for two power products, providing us with a notion of relative size for power products. We can then try to change an ideal basis in a systematic way to create one with polynomials having terms a_iP_i with as "small" power products P_i as possible. We denote by 1 the power product with all exponents 0, and require that an ordering of the power products has the properties shown in the box. Suppose that such an ordering has been described and that $P_i \neq P_j$ and P_i divides P_j so that $P_j = PP_i$ where $1 < P$. From Property 4 in the box, we then have $1P_i < PP_i = P_j$, so $P_i < P_j$. Thus P_i divides P_j implies that $P_i < P_j$. In Exercise 28, we ask you to show by a counterexample that $P_i < P_j$ does not imply that P_i divides P_j.

It can also be shown that these properties guarantee that any step-by-step process for modifying a finite ideal basis that does not increase the size of any maximal power product in a basis element and replaces at least one by something smaller at each step will terminate in a finite number of steps.

In $F[x]$ with x the only indeterminate, there is only one power product ordering, for by Property 1, we must have $1 < x$. Multiplying repeatedly by x and using Property 4, we have $x < x^2$, $x^2 < x^3$, etc. Property 3 then shows that $1 < x < x^2 < x^3 < \cdots$ is the only possible order. Notice that in Example 28.8, we modified a basis by replacing basis polynomials by polynomials containing smaller power products.

There are a number of possible orderings for power products in $F[\mathbf{x}]$ with n indeterminates. We present just one, the *lexicographical order* (denoted by "lex"). In lex, we define

$$x_1^{s_1}x_2^{s_2}\cdots x_n^{s_n} < x_1^{t_1}x_2^{t_2}\cdots x_n^{t_n} \tag{2}$$

if and only if $s_i < t_i$ for the first subscript i, reading from left to right, such that $s_i \neq t_i$. Thus in $F[x, y]$, if we write power products in the order $x^n y^m$, we have $y = x^0y^1 < x^1y^0 = x$ and $xy < xy^2$. Using lex, the order of n indeterminates is given by $1 < x_n < x_{n-1} < \cdots < x_2 < x_1$. Our reduction in Example 28.7, where we first got rid of all "big" x's that we could and then the "smaller" y's, corresponded to the lex order $z < y < x$, that is, to writing all power products in the $x^m y^n z^s$ order. For the two-indeterminate case with $y < x$, the total lex term order schematically is

$$1 < y < y^2 < y^3 \cdots < x < xy < xy^2 < xy^3 < \cdots < x^2 < x^2y < x^2y^2 < \cdots.$$

An ordering of power products P induces an obvious ordering of terms aP of a polynomial in $F[\mathbf{x}]$, which we will refer to as a **term order.** From now on, given an ordering of power products, we consider every polynomial f in $F[\mathbf{x}]$ to be written in decreasing order of terms, so that the leading (first) term has the highest order. We denote by $\mathrm{lt}(f)$ the leading term of f and by $\mathrm{lp}(f)$ the power product of the leading term. If f and g are polynomials in $F[\mathbf{x}]$ such that $\mathrm{lp}(g)$ divides $\mathrm{lp}(f)$, then we can execute a division of f by g, as illustrated in the linear and one-indeterminate cases, to obtain $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$ where $\mathrm{lp}(r) < \mathrm{lp}(f)$. Note that we did not say that $\mathrm{lp}(r) < \mathrm{lp}(g)$. We illustrate with an example.

28.9 Example By division, reduce the basis $\{xy^2, y^2 - y\}$ for the ideal $I = \langle xy^2, y^2 - y\rangle$ in $\mathbb{R}[x, y]$ to one with smaller maximum term size, assuming the order lex with $y < x$.

Solution We see that y^2 divides xy^2 and compute

$$\begin{array}{r|l} & x \\ \hline y^2 - y & xy^2 \\ & xy^2 - xy \\ \hline & xy \end{array}$$

Because y^2 does not divide xy, we cannot continue the division. Note that $\mathrm{lp}(xy) = xy$ is not less than $\mathrm{lp}(y^2 - y) = y^2$. However, we do have $\mathrm{lp}(xy) < \mathrm{lp}(xy^2)$. Our new basis for I is $\{xy, y^2 - y\}$. ▲

When dealing with more than one indeterminate, it is often easier to perform basis reduction by multiplying a basis polynomial $g(\mathbf{x})$ by a polynomial $-q(\mathbf{x})$ and adding it to a polynomial $f(\mathbf{x})$ to obtain $r(\mathbf{x})$, as we perform matrix reduction in linear algebra, rather than writing out the division display as we did in the preceding example. Starting with basis polynomials xy^2 and $y^2 - y$, we can reduce the xy^2 by multiplying $y^2 - y$ by $-x$ and adding the resulting $-xy^2 + xy$ to xy^2, obtaining the replacement xy for xy^2. We can do that in our head, and write down the result directly.

Referring again to Example 28.9, it will follow from what we state later that given any polynomial $f(x, y) = c_1(x, y)(xy) + c_2(x, y)(y^2 - y)$ in $\langle xy, y^2 - y\rangle$, either xy or

y^2 will divide $\mathrm{lp}(f)$. (See Exercises 31.) This illustrates the defining property of a *Gröbner basis*.

28.10 Definition A set $\{g_1, g_2, \cdots, g_r\}$ of nonzero polynomials in $F[x_1, x_2, \cdots, x_n]$, with term ordering $<$, is a **Gröbner basis** for the ideal $I = \langle g_1, g_2, \cdots, g_r \rangle$ if and only if, for each nonzero $f \in I$, there exists some i where $1 \leq i \leq r$ such that $\mathrm{lp}(g_i)$ divides $\mathrm{lp}(f)$. ■

While we have illustrated the computation of a Gröbner basis from a given basis for an ideal in Examples 28.7, 28.8, and 28.9, we have not given a specific algorithm. We refer the reader to Adams and Loustaunau [23]. The method consists of multiplying some polynomial in the basis by any polynomial in $F[\mathbf{x}]$ and adding the result to another polynomial in the basis in a manner that reduces the size of power products. In our illustrations, we have treated the case involving division of $f(\mathbf{x})$ by $g(\mathbf{x})$ where $\mathrm{lp}(g)$ divides $\mathrm{lp}(f)$, but we can also use the process if $\mathrm{lp}(g)$ only divides some other power product in f. For example, if two elements in a basis are $xy - y^3$ and $y^2 - 1$, we can multiply $y^2 - 1$ by y and add it to $xy - y^3$, reducing $xy - y^3$ to $xy - y$. Theorem 28.6 shows that this is a valid computation.

You may wonder how any basis $\{g_1, g_2, \cdots, g_r\}$ can fail to be a Gröbner basis for $I = \langle g_1, g_2, \cdots, g_r \rangle$ because, when we form an element $c_1 g_1 + c_2 g_2 + \cdots + c_r g_r$ in I, we see that $\mathrm{lp}(g_i)$ is a divisor of $\mathrm{lp}(c_i g_i)$ for $i = 1, 2, \cdots, r$. However, cancellation of power products can occur in the addition. We illustrate with an example.

28.11 Example Consider the ideal $I = \langle x^2y - 2, xy^2 - y \rangle$ in $\mathbb{R}[x, y]$. The polynomials in the basis shown cannot be reduced further. However, the ideal I contains $y(x^2y - 2) - x(xy^2 - y) = xy - 2y$, whose leading power product xy is not divisible by either of the leading power products x^2y or xy^2 of the given basis. Thus $\{x^2y - 2, xy^2 - y\}$ is not a Gröbner basis for I, according to Definition 28.10. ▲

When we run into a situation like that in Example 28.11, we realize that a Gröbner basis must contain some polynomial with a smaller leading power product than those in the given basis. Let f and g be polynomials in the given basis. Just as we did in Example 28.11, we can multiply f and g by as small power products as possible so that the resulting two leading power products will be the same, the *least common multiple* (lcm) of $\mathrm{lp}(f)$ and $\mathrm{lp}(g)$, and then subtract or add with suitable coefficients from F so cancellation results. We denote a polynomial formed in this fashion by $S(f, g)$. We state without proof a theorem that can be used to test whether a basis is a Gröbner basis.

28.12 Theorem A basis $G = \{g_1, g_2, \cdots, g_r\}$ is a Gröbner basis for the ideal $\langle g_1, g_2, \cdots, g_r \rangle$ if and only if, for all $i \neq j$, the polynomial $S(g_i, g_j)$ can be reduced to zero by repeatedly dividing remainders by elements of G, as in the division algorithm.

As we mentioned before, we may prefer to think of reducing $S(g_i, g_j)$ by a sequence of operations consisting of adding (or subtracting) multiples of polynomials in G, rather than writing out division.

We can now indicate how we can obtain a Gröbner basis from a given basis. First, reduce the polynomials in the basis as far as possible among themselves. Then choose

polynomials g_i and g_j in the basis, and form the polynomial $S(g_i, g_j)$. See if $S(g_i, g_j)$ can be reduced to zero as just described. If so, choose a different pair of polynomials, and repeat the procedure with them. If $S(g_i, g_j)$ cannot be reduced to zero as described above, augment the given basis with this $S(g_i, g_j)$, and start all over, reducing this basis as much as possible. By Theorem 28.12, when every polynomial $S(g_i, g_j)$ for all $i \neq j$ can be reduced to zero using polynomials from the latest basis, we have arrived at a Gröbner basis. We conclude with a continuation of Example 28.11.

28.13 Example Continuing Example 2.8.11, let $g_1 = x^2y - 2$, $g_2 = xy^2 - y$, and $I = \langle g_1, g_2 \rangle$ in $\mathbb{R}^2$. In Example 28.11, we obtained the polynomial $S(g_1, g_2) = xy - 2y$, which cannot be reduced to zero using g_1 and g_2. We now reduce the basis $\{x^2y - 2, xy^2 - y, xy - 2y\}$, indicating each step.

$\{x^2y - 2, xy^2 - y, xy - 2y\}$	augmented basis
$\{2xy - 2, xy^2 - y, xy - 2y\}$	by adding $(-x)$ (third) to first
$\{2xy - 2, 2y^2 - y, xy - 2y\}$	by adding $(-y)$ (third) to second
$\{4y - 2, 2y^2 - y, xy - 2y\}$	by adding (-2) (third) to first
$\{4y - 2, 0, xy - 2y\}$	by adding $(-\frac{y}{2})$ (first) to second
$\{4y - 2, 0, \frac{1}{2}x - 2y\}$	by adding $(-\frac{x}{4})$ (first) to third
$\{4y - 2, 0, \frac{1}{2}x - 1\}$	by adding $(\frac{1}{2})$ (first) to third

Clearly, $\{y - \frac{1}{2}, x - 2\}$ is a Gröbner basis. Note that if $f = y - \frac{1}{2}$ and $g = x - 2$, then $S(f, g) = xf - yg = (xy - \frac{x}{2}) - (xy - 2y) = -\frac{x}{2} + 2y$, which can readily be reduced to zero by adding $\frac{1}{2}(x - 2)$ and $-2(y - \frac{1}{2})$.

From the Gröbner basis, we see that the algebraic variety $V(I)$ contains only one point, $(2, \frac{1}{2})$, in $\mathbb{R}^2$. ▲

The importance of Gröbner bases in applications is due to the fact that they are *machine computable*. They have applications to engineering and computer science as well as to mathematics.

■ EXERCISES 28

In Exercises 1 through 4, write the polynomials in $\mathbb{R}[x, y, z]$ in decreasing term order, using the order lex for power products $x^m y^n z^s$ where $z < y < x$.

1. $2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3$

2. $3y^2z^5 - 4x + 5y^3z^3 - 8z^7$

3. $3y - 7x + 10z^3 - 2xy^2z^2 + 2x^2yz^2$

4. $38 - 4xz + 2yz - 8xy + 3yz^3$

In Exercises 5 through 8, write the polynomials in $\mathbb{R}[x, y, z]$ in decreasing term order, using the order lex for power products $z^m y^n x^s$ where $x < y < z$.

5. The polynomial in Exercise 1.

6. The polynomial in Exercise 2.

7. The polynomial in Exercise 3.

8. The polynomial in Exercise 4.

Another ordering, deglex, for power products in $F[\mathbf{x}]$ is defined as follows:

$$x_1^{s_1}x_2^{s_2}\cdots x_n^{s_n} < x_1^{t_1}x_2^{t_2}\cdots x_n^{t_n}$$

if and only if either $\sum_{i=1}^{n} s_i < \sum_{i=1}^{n} t_i$, or these two sums are equal and $s_i < t_i$ for the smallest value of i such that $s_i \neq t_i$. Exercises 9 through 13 are concerned with the order deglex.

9. List, in increasing order, the smallest 20 power products in $\mathbb{R}[x, y, z]$ for the order deglex with power products $x^m y^n z^s$ where $z < y < x$.

In Exercises 10 through 13, write the polynomials in order of decreasing terms using the order deglex with power products $x^m y^n z^s$ where $z < y < x$.

10. The polynomial in Exercise 1.

11. The polynomial in Exercise 2.

12. The polynomial in Exercise 3.

13. The polynomial in Exercise 4.

For Exercises 14 through 17, let power products in $\mathbb{R}[x, y, z]$ have order lex where $z < y < x$. If possible, perform a single-step division algorithm reduction that changes the given ideal basis to one having smaller maximum term order.

14. $\langle xy^2 - 2x, x^2y + 4xy, xy - y^2 \rangle$

15. $\langle xy + y^3, y^3 + z, x - y^4 \rangle$

16. $\langle xyz - 3z^2, x^3 + y^2z^3, x^2yz^3 + 4 \rangle$

17. $\langle y^2z^3 + 3, y^3z^2 - 2z, y^2z^2 + 3 \rangle$

In Exercises 18 and 19, let the order of power products in $\mathbb{R}[w, x, y, z]$ be lex with $z < y < x < w$. Find a Gröbner basis for the given ideal.

18. $\langle w + x - y + 4z - 3, 2w + x + y - 2z + 4, w + 3x - 3y + z - 5 \rangle$

19. $\langle w - 4x + 3y - z + 2, 2w - 2x + y - 2z + 5, w - 10x + 8y - z - 1 \rangle$

In Exercises 20 through 22, find a Gröbner basis for the indicated ideal in $\mathbb{R}[x]$.

20. $\langle x^4 + x^3 - 3x^2 - 4x - 4, x^3 + x^2 - 4x - 4 \rangle$

21. $\langle x^4 - 4x^3 + 5x^2 - 2x, x^3 - x^2 - 4x + 4, x^3 - 3x + 2 \rangle$

22. $\langle x^5 + x^2 + 2x - 5, x^3 - x^2 + x - 1 \rangle$

In Exercises 23 through 26, find a Gröbner basis for the given ideal in $\mathbb{R}[x, y]$. Consider the order of power products to be lex with $y < x$. If you can, describe the corresponding algebraic variety in $\mathbb{R}[x, y]$.

23. $\langle x^2y - x - 2, xy + 2y - 9 \rangle$

24. $\langle x^2y + x, xy^2 - y \rangle$

25. $\langle x^2y + x + 1, xy^2 + y - 1 \rangle$

26. $\langle x^2y + xy^2, xy - x \rangle$

Concepts

27. Let F be a field. Mark each of the following true or false.

_____ **a.** Every ideal in $F[\mathbf{x}]$ has a finite basis.

_____ **b.** Every subset of $\mathbb{R}^2$ is an algebraic variety.

_____ **c.** The empty subset of $\mathbb{R}^2$ is an algebraic variety.

_____ **d.** Every finite subset of $\mathbb{R}^2$ is an algebraic variety.

_____ **e.** Every line in $\mathbb{R}^2$ is an algebraic variety.

_____ **f.** Every finite collection of lines in $\mathbb{R}^2$ is an algebraic variety.

_____ **g.** A greatest common divisor of a finite number of polynomials in $\mathbb{R}[x]$ (one indeterminate) can be computed using the division algorithm repeatedly.

_____ **h.** I have computed Gröbner bases before I knew what they were.

_____ **i.** Any ideal in $F[\mathbf{x}]$ has a unique Gröbner basis.

_____ **j.** The ideals $\langle x, y \rangle$ and $\langle x^2, y^2 \rangle$ are equal because they both yield the same algebraic variety, namely $\{(0, 0)\}$, in $\mathbb{R}^2$.

28. Let $\mathbb{R}[x, y]$ be ordered by lex. Give an example to show that $P_i < P_j$ does not imply that P_i divides P_j.

Theory

29. Show that if $f_1, f_2, \cdots, f_r$ are elements of a commutative ring R with unity, then $I = \{c_1 f_1 + c_2 f_2 + \cdots + c_r f_r \mid c_i \in I \text{ for } i = 1, \cdots, r\}$ is an ideal of R.

30. Show that if $f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x})$ in $F[\mathbf{x}]$, then the common divisors in $F[\mathbf{x}]$ of $f(\mathbf{x})$ and $g(\mathbf{x})$ are the same as the common divisors in $F[\mathbf{x}]$ of $g(\mathbf{x})$ and $r(\mathbf{x})$.

31. Show that $\{xy, y^2 - y\}$ is a Gröbner basis for $\langle xy, y^2 - y\rangle$, as asserted after Example 28.9.

32. Let F be a field. Show that if S is a nonempty subset of F^n, then

$$I(S) = \{f(\mathbf{x}) \in F[\mathbf{x}] \mid f(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}$$

is an ideal of $F[\mathbf{x}]$.

33. Referring to Exercise 32, show that $S \subseteq V(I(S))$.

34. Referring to Exercise 32, give an example of a subset S of $\mathbb{R}^2$ such that $V(I(S)) \neq S$.

35. Referring to Exercise 32, show that if N is an ideal of $F[\mathbf{x}]$, then $N \subseteq I(V(N))$.

36. Referring to Exercise 32, give an example of an ideal N in $\mathbb{R}[x, y]$ such that $I(V(N)) \neq N$.